



Indian Institute of Technology, Kanpur

Department of Statistics & Data Science

PLACEMENT BROCHURE 2026-27

Master of Science in Statistics



Homepage
Department of Statistics & Data Science

Website: <https://iitk.ac.in/sds>
Phone: +91-512-259 4433



Homepage,
Students' Placement Office

■ Welcome Message



Prof. Neeraj Misra

Head of the Department
head_sds@iitk.ac.in

The Department of Statistics and Data Science at IIT Kanpur, carrying forward a distinguished academic legacy established in 1970, is a premier center for education in statistics, data science, machine learning, and artificial intelligence. Formerly part of the Department of Mathematics and Statistics, it now stands independently to focus on statistical theory and computational methods. Our BS in Statistics and Data Science and MSc in Statistics offer rigorous theoretical and applied training, preparing students for impactful careers in academia, industry, and research.

Our faculty are internationally recognized experts actively engaged in cutting-edge research, and they bring this expertise into the classroom. The department fosters a vibrant academic environment through regular seminars, workshops, and collaborative projects.

Our graduates have consistently performed well in placements and are highly sought after by top institutions and organisations worldwide.

“ We warmly welcome recruiters to connect with our students through the campus placement program at IIT Kanpur.

■ About the Department

The Department of Statistics & Data Science at IIT Kanpur, is a leading centre for both pure and applied statistical research. It combines deep theoretical foundations with practical applications in statistics, machine learning, and data science. The department boasts distinguished faculty, active research output, and strong global collaborations, especially in Probability and Statistics. It ranked 36th globally in AI & Data Science in 2024, reflecting its excellence in modern computational fields. Offering programs at all academic levels, it prepares students for careers in academia, research, and industry. Its curriculum is supported by interdisciplinary projects and strong industry ties.

■ About the Programs

M.Sc. in Statistics (2-Year):

Objective: Cultivate a rigorous mathematical and statistical mindset, foster research interests, and train computational statisticians.

Program Structure:

Year I: Foundational Coursework — Probability, Inference, Real/Complex Analysis, Stochastic Processes, Data Science Lab.

Year II: Advanced Coursework — Multivariate Analysis, Regression, Bayesian Methods, Time Series.

Applied Learning: Hands-on laboratory work, semester-long projects, and specialized electives covering simulation techniques, artificial intelligence, machine learning, and other modern data science methodologies.

Distinguished Faculty

The department takes pride in its distinguished faculty, which forms the foundation of our educational excellence. With a total of 11 professors, assistant professors, and associate professors, our faculty's diverse research interests span both pure and applied mathematics and statistics, enriching the academic environment and fostering innovation.

DR. AMIT MITRA (PhD, IIT Kanpur)
Professor
E-mail: amitra@iitk.ac.in

DR. ARNAB HAZRA (PhD, NC State University, Raleigh, US)
Assistant Professor
E-mail: ahazra@iitk.ac.in

DR. SHARMISHTHA MITRA (PhD, IIT Kanpur)
Professor
E-mail: smitra@iitk.ac.in

DR. NEERAJ MISRA (PhD, IIT Kanpur)
Professor
E-mail: neeraj@iitk.ac.in

DR. SUPRIO BHAR (PhD, ISI Bangalore)
Assistant Professor
E-mail: suprio@iitk.ac.in

DR. SATYA PRAKASH SINGH (PhD, IIT Bombay)
Assistant Professor
E-mail: singhsp@iitk.ac.in

DR. SHALABH (PhD, University of Lucknow)
Professor
E-mail: shalab@iitk.ac.in

DR. DEBASIS KUNDU (PhD, Pennsylvania State University)
Professor
E-mail: kundu@iitk.ac.in

DR. SOUMYARUP SADHUKHAN (PhD, ISI Kolkata)
Assistant Professor
E-mail: soumyarups@iitk.ac.in

DR. DOOTIKA VATS (PhD, University of Minnesota)
Associate Professor
E-mail: dootika@iitk.ac.in

DR. SUBHRA SANKAR DHAR (PhD, ISI Kolkata)
Professor
E-mail: subhra@iitk.ac.in

■ Computing Skills

Technical Skills

- Languages: Python, R, SQL, C++
- Python Libraries: Numpy, Pandas, Scikit-Learn, SciPy, Matplotlib, Seaborn, statsmodels, PyTorch, TensorFlow
- R Packages: ggplot2, dplyr, Rcpp, shiny, rvest, stats, tidyverse
- Utilities: MS Office, LaTeX, R Markdown, AI Tools, GitHub



Analytical & Modeling Skills

- Machine Learning (ML) & AI – classification, clustering, and predictive modeling
- Bayesian Models & MCMC – probabilistic inference and uncertainty estimation
- Computing Skills – R, Python, data manipulation, numerical optimization, and Monte Carlo simulations
- Time Series Analysis – forecasting and trend modeling
- Shiny Apps & Web Scraping – interactive data dashboards and real-world data acquisition

Courses Undertaken

Core Courses

- Probability Theory
- Sampling Theory
- Stochastic Processes
- Statistical Computing
- Time Series Analysis
- Linear Regression & ANOVA
- Statistical Inference
- Bayesian Analysis
- Multivariate Analysis
- Real Analysis
- Complex Analysis
- Matrix Algebra
- Linear Estimation
- Data Science Lab 1
- Data Science Lab 2
- Data Science Lab 3

Department Electives

- Statistical Pattern Recognition
- Markov Chain Monte Carlo
- Spatial Statistics
- ANN/ML Approach for Differential Equations
- Statistical Decision Theory
- Robust Statistical Methods
- Order Statistics
- Econometrics
- Nonlinear Regression
- Statistical Reliability Theory
- Data Mining
- Non-Parametric Inference
- Empirical Processes
- Set Theory & Discrete Mathematics
- Ordinary Differential Equations
- Partial Differential Equations
- Measure Theory
- Game Theory

Institute Open Electives

- Parallel Computing
- Parallel Programming
- Computer Vision and Image Processing
- Big Data Analysis
- Probabilistic Modelling and Inference
- Introduction to Machine Learning
- Probabilistic Machine Learning
- Statistical Natural Language Processing
- Statistical Signal Processing
- Digital Image Processing
- Convex Optimization in SP/COM
- Detection and Estimation Theory
- Speech Signal Processing
- Computational Intelligence for Machine Vision, Automation and Control
- Neural Networks
- Machine Learning for Signal Processing
- Optimization for Big Data
- Artificial Intelligence, Machine Learning and its Applications
- Advanced Topics in Machine Learning
- Principles of Data Base Systems
- Embedding and Cyber-Physical Systems
- Theory of Computation
- Bayesian Econometrics
- Advanced Algorithms
- Randomized Algorithms
- Computer Organisation
- Software Engineering
- Computational Complexity
- Computational Number Theory
- Modern Cryptology
- Data Mining

Areas of Expertise

Theoretical Foundations

- **Real Analysis** – underpinning rigorous algorithm design and advanced statistical modelling.
- **Probability & Inference** – essential for uncertainty quantification, predictive risk models, reliability analysis.
- **Markov Chains & Stochastic Processes** – key to simulation, time-series forecasting, financial mathematics and queuing systems.

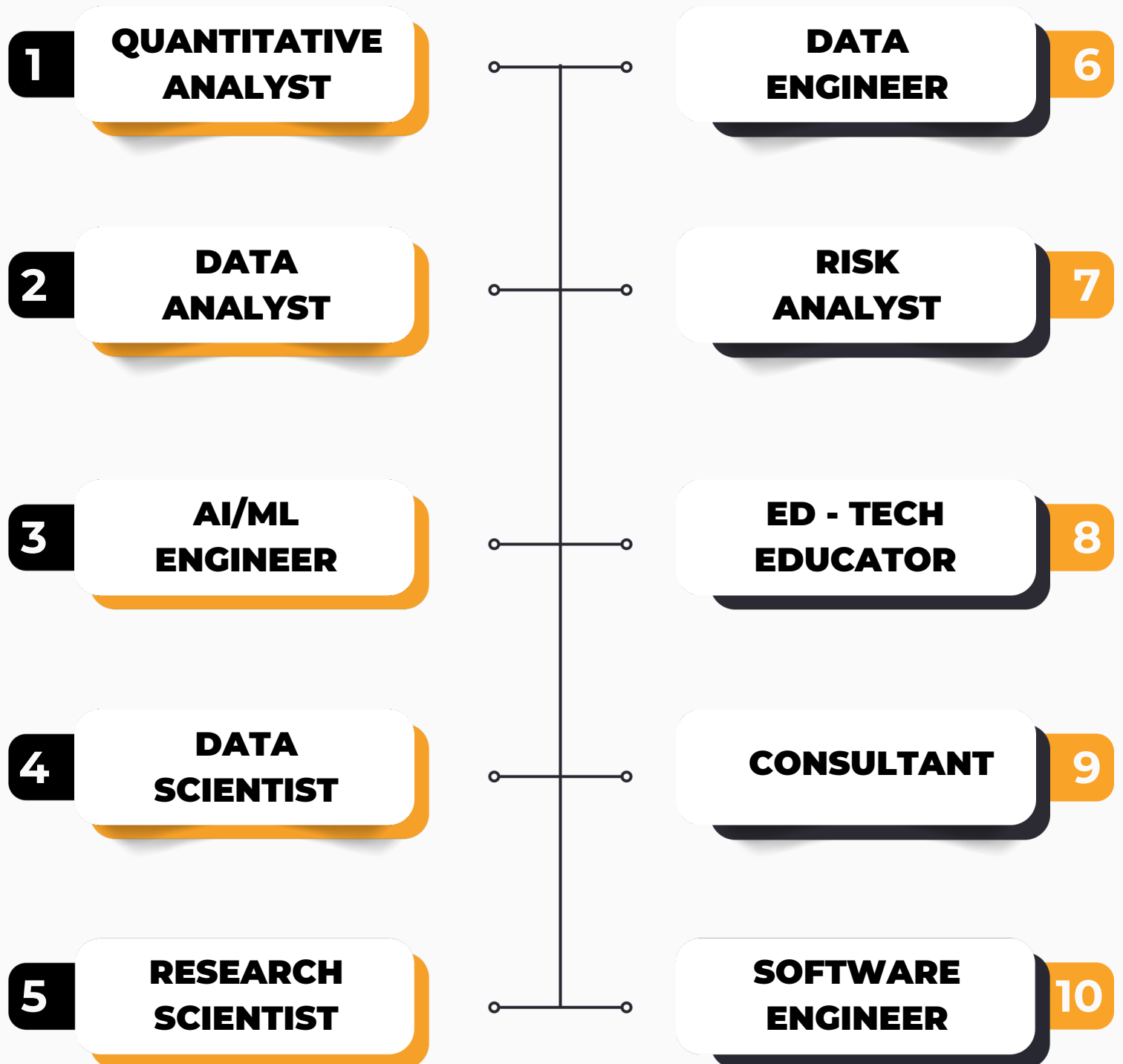
Industry-Focused Applications

- **Natural Language Processing (NLP)** – text analytics using sequence models and deep learning pipelines.
- **Econometrics & Financial Analytics** – predictive econometric modelling, trading strategies, and risk forecasting.
- **Bioinformatics & Health Analytics** – modelling genomics, clinical and health-related data using statistical pipelines.
- **Computer Vision** – applied image/data analysis for diagnostics, surveillance, and automated processing.

Advanced Methodologies & Techniques

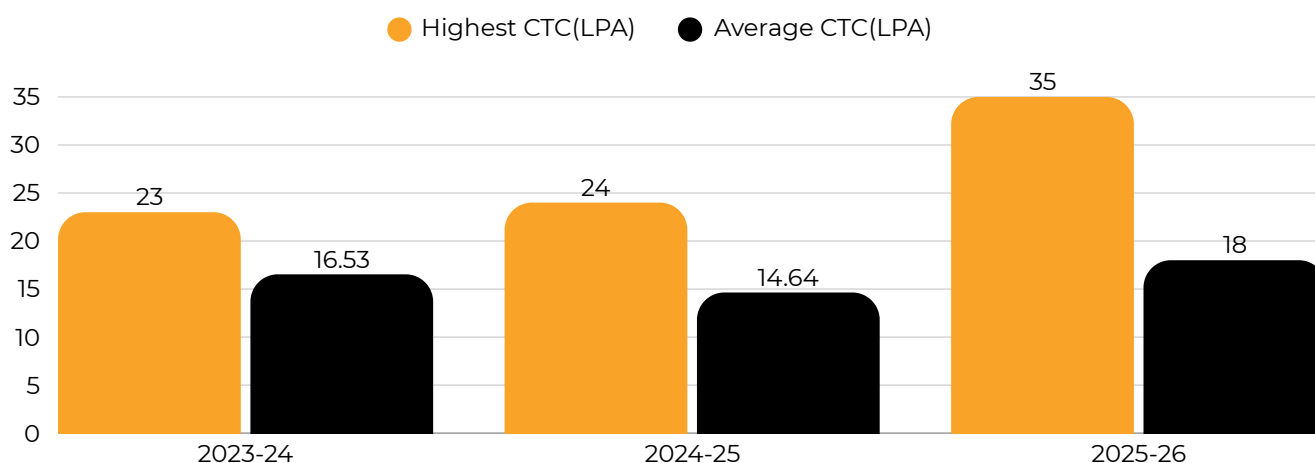
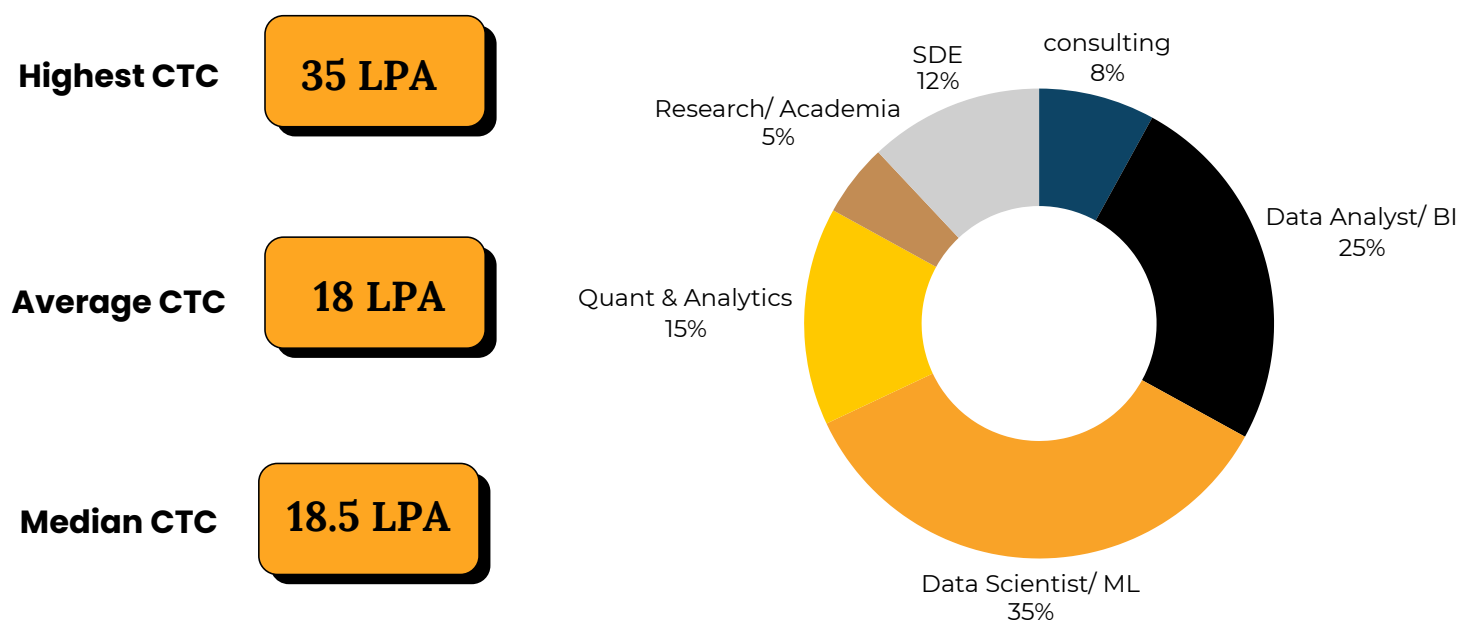
- **Clustering, Neural Networks, SGD** – unsupervised learning and deep neural approaches for segmentation, classification, pattern detection.
- **Markov Chain Monte Carlo (MCMC)** – Bayesian inference methods used for simulations and probabilistic modelling.
- **Bayesian Modelling** – Hierarchical Bayesian models and probabilistic inference for uncertainty-aware and interpretable modelling.
- **Spatial Statistics** – Spatial statistical inference and kriging, with applications in ground station-based and satellite-based data modeling and analysis.

Profiles Offered



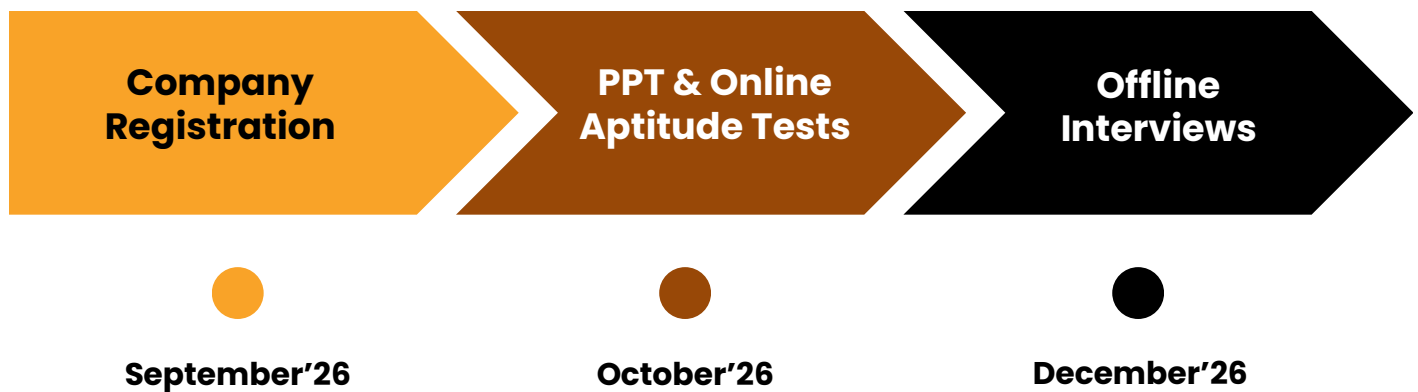
Placement Statistics

Packages and Profiles Offered in 2025-26



The M.Sc. Statistics cohort witnessed an exceptional placement season, driven by top-tier roles spanning AI Decision Science, Data Science, Biostatistics, Quantitative Analytics, and Corporate Management. Leading global firms and financial institutions heavily recruited our graduates, reflecting the rigorous analytical training and industry readiness of the program.

■ Placement Timeline 2026-27



1st Week of Sept 2026: Companies can register on IITK's Recruitment Automation System (RAS) portal and share Job Descriptions by filling out the Job Announcement Form (JAF).

Oct - Nov 2026: Companies conduct their Pre-Placement Taks (PPTs) and tests or Screening process after finalizing the schedule with the placement team.

1st Week of Dec 2026: The placement office allots dates and slots of interviews and PoC for conducting in-person interviews of the shortlisted candidates on campus.



Past Recruiters



Thank You!

Let's Get
In Touch



Contact Us

Faculty Placement Convenor

Prof. Arnab Hazra
Phone: +91-8337065993
E-mail: ahazra@iitk.ac.in

Department Placement Coordinators

Ayush Kundu
Phone: +91-9674016704
E-mail: ayushkundu25@iitk.ac.in

Kriti Maheshwari
Phone: +91-9027574782
E-mail: kmaheshwari25@iitk.ac.in